

Libraries required

The following libraries must either be installed globally in your python directory (i.e. anaconda, miniconda) or in the python environment being used to use srimg. This should be done automatically when installing srimg, but can be done manually if for some reason the dependencies are not properly installed. These libraries can be installed using the listed commands, either from Conda, PyPi, or docker.

- Numpy
 - `conda install numpy`
 - `pip install numpy`
- Pandas
 - `conda install pandas`
 - `pip install pandas`
- Openpyxl
 - `conda install anaconda::openpyxl`
- Pysrim
 - `conda install -c costrouc pysrim`
 - `pip install pysrim`
 - `docker pull costrouc/pysrim`

Modified pysrim files to replace

All srimg functions use the pysrim library to read SRIM output files before performing calculations. However, the latest version of this library is not compatible with the python versions 3.12 and beyond. The srimg repository contains updated versions of certain pysrim files that make it compatible with python 3.14 (the latest version of python as of the release of srimg), with additional minor modifications to allow for increased computational efficiency in srimg. These incompatibilities with current versions of python were pointed out by comments posted in the Issues tab of the pysrim GitHub. Specifically, the the yaml loader issue was fixed, and the use of distutils was replaced by shutils. These files are available in the srimg repository in `srimg/pysrim_updates`. After installing pysrim, the following instructions must be followed to replace the outdated pysrim files.

1. Download the three files within `srimg/pysrim_updates`, these will be used to replace their corresponding pysrim files
2. Using Windows Powershell, Anaconda Prompt, or equivalent, run the command `pip show pysrim` to find the exact directory in which pysrim is installed
3. Copy/cut the pysrim_updates files, then navigate to this directory (the *srimg* folder within the directory given by the previous step) in file explorer and paste the copied files into the srimg folder
4. Delete the original versions of these files from pysrim
 - a. In the main srimg folder, delete `output.py`

- b. In the main `srin` folder, delete `srin.py`
 - c. In the `core` subfolder, delete `elementdb.py`
- 5. Rename the `pysrin_updated` files to the names of their corresponding `pysrin` files that were just deleted (i.e. remove the “_updated” from the end of the name of each file)
- 6. Move the new `elementdb.py` file to the `core` folder